
MC311 Mathematical Optimization (3-1-0-4)
Autumn 2025

1. **Instructor:** Aditya Tatu
2. **Course Category:** Core for B.Tech (Mathematics & Computing)
3. **Prerequisite Concepts:** *Calculus* - Taylor's series, Taylor's remainder theorem, Continuity and Differentiability of functions of one and several variables, Gradient of a function, Jacobian matrix, Hessian matrix of a function. *Linear/Matrix Algebra* - Rank of a matrix, Column space & Nullspace, Eigenvalues and Eigenvectors, Matrix Diagonalization, Norms and Inner products in $\mathbb{R}^n$.
4. **Course Summary:** A lot of problems across engineering and management can be modeled as optimization problems. Optimization problems come in various forms - Continuous and Discrete or Combinatorial Optimization being two of the most popular ones. This course deals exclusively with Continuous optimization, and aims to introduce both, theory and algorithms in Continuous optimization. The course will introduce students to basic terminology in optimization, theory - conditions for existence of points of optima(maxima/minima), and algorithms to solve optimization problems. We will deal with both, linear and non-linear continuous optimization problems, focusing more on the latter.
5. **Course Content:**
 - (a) Basic terminology in Optimization: Local optima, Global Optima, Equivalence of minimization and maximization, Convex sets and functions.
 - (b) Set Constrained/Unconstrained optimization theory: First and Second order Necessary/Sufficient conditions for optima.
 - (c) Unconstrained optimization algorithms: Gradient Methods, Newton's method, Conjugate direction methods.
 - (d) Constrained non-linear optimization theory: Feasibility set and surface, Active and inactive conditions, Lagrange multiplier, Karush-Kuhn-Tucker(KKT) condition.
 - (e) Special cases of Constrained Optimization: Linear programming, Basic feasible solutions, Simplex method.
 - (f) Duality: Lagrangian duality, Geometric interpretation, Duality gap, Saddle point interpretation.
 - (g) (If time permits) Constrained Optimization algorithms: Barrier & Penalty methods
6. **References:**
 - (a) An Introduction to Optimization, Chong and Zak, Fourth ed., Wiley.
 - (b) Linear and Nonlinear Programming, Luenberger and Ye, Fifth ed., Springer.

(c) Convex Optimization, Boyd and Vandenberghe, Cambridge University Press.

7. Course Outcomes: The student will be able to

- (a) use the first and second order conditions to compute optima for *small* unconstrained optimization problems,
- (b) implement line search, gradient descent, and Newton's method for unconstrained optimization,
- (c) use the KKT conditions to derive conditions that an optima or a constrained non-linear optimization problem has to satisfy,
- (d) appreciate importance of optimization in Data Science applications.

8. Evaluation:

- (a) Quizzes, Assignments/Lab exam - 30 – 40%.
- (b) In Sem 1 & 2 and/or Project - 30 – 40%.
- (c) End Sem - 30 – 40%.

9. Program Outcomes:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X									